



Lost and found

NASA's Spitzer space telescope spotted some tell-tale infrared background radiation that could indicate the presence of ancient stars

Hybrid capsule

NASA is currently testing space capsules with attached helicopter wings for better descent control

harvesters each process millions of tonnes of regolith every year, to produce thousands of tonnes of Helium 3, which is then shipped off to Earth via a mass driver and orbital collector mechanism. The entire process is entirely automated and the harvesters follow very precise paths that are monitored remotely.

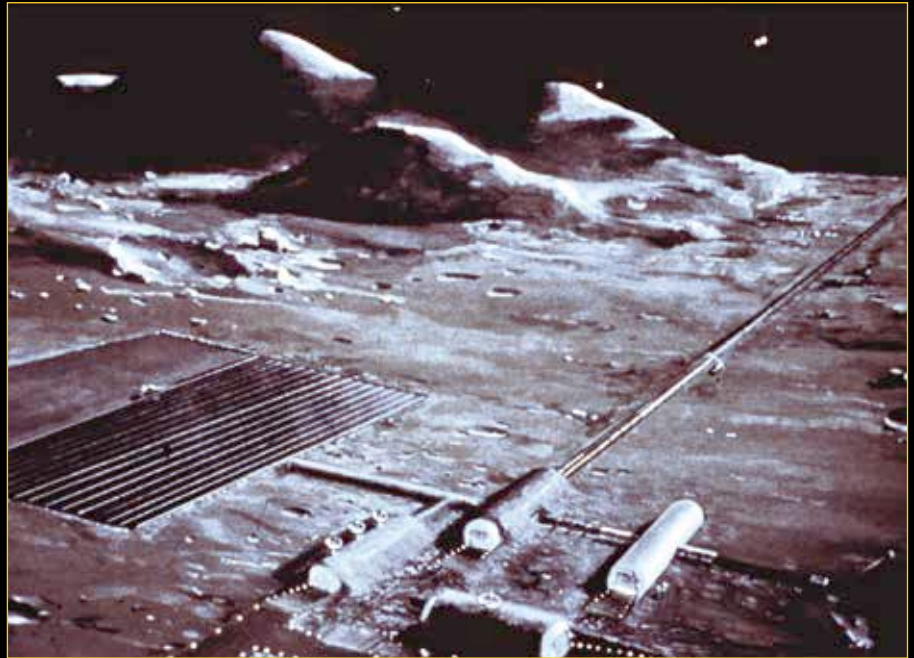
The entire, partly processed ore is wrapped up in a large, plastic bag of sorts before being sent off to the "magazine". The magazine is where the ore parcels (called pellets) are wrapped in foil and then loaded into the mass driver. The mass driver, powered by the energy from vast solar arrays on the poles, is a feat of engineering in itself. With a total build cost of over \$18 billion, it is the single most expensive cannon ever produced. The device uses magnetic levitation and accelerates the package to mach 25 (1 mach measured as the speed of sound at sea level on Earth in this case). Since this is the moon and escape velocities are so low, the package can easily escape the moon's gravitational force and is launched into a High Earth Orbit.

Ships from the one of the 10 IOPs will rendezvous with the packages at regular intervals, usually at weekly intervals, where they will round up the packages one after the other, fit them with parachutes and retro-boosters and return to the IOP. The boosters will then be triggered remotely and will guide the packages back to Earth for reentry. The precision required and, actually achieved, is quite staggering, with only 1 in 10,000 packages missing their designated drop-zones by more than 2 miles. These packages are almost priceless in value, providing clean energy for the Earth, each package capable of running a country the size of UK for a few weeks at a time.

After the package is retrieved, the boosters are inspected, refuelled, and launched to one of the IOPs for reuse. A very energy efficient process considering the elaborate process that is involved.

Day 2: Hydroponic farms

The second day will involve a trip to the moon's hydroponic farms that are present near the poles of the moon. The journey will be a long one and will involve a maglev



A lunar mass driver capable of incredible acceleration in the low gravity

train ride that will take a few hours. All facilities, including entertainment will be provided on the train so you will have quite a relaxed and pleasant journey.

LunaCorp farms

Hydroponic farms were the only and inevitable, viable farming option on the moon. Given the scarcity of water on the lunar surface, this was the only method that could conserve and recycle water to the degree that we would like. This method of farming involves no soil and barely any water, which is absolutely perfect for the limited resources available on the moon. The poles were the obvious choice given that that was where all the water was (frozen though it may be), and also where the day night cycles at least resembled the Earth's, in part. The food production has been refined over the last few decades and long gone are the days when astronauts had to depend on concoctions made from green algae and now you can get everything from potatoes to rice and wheat and even fruits. The food isn't that delectable of course, but that is more to do with the low gravity messing with your taste buds than anything else.

The food in the hotel is shipped directly from Earth though. As you would have

noticed, it's primarily a vegan diet here. Hardly appealing to everyone's palate wouldn't you agree?

Day 3: LunarSat

Day 3 will take you on a tour of the observatories setup on the moon, again, these are at the poles. Each telescope costs over a billion dollars, but these, along with the vast arrays of radio telescopes, are some of the most advanced ever devised by man. The almost negligible atmosphere and the relatively fixed position of the poles are absolutely ideal for peeking into the dark recesses of space. The scientists here are extremely possessive about their instruments, with good reason of course, so you will only be allowed to visit the science centre and will be given a tour of the facilities from the outside. These telescopes and radars are not just meant for space exploration. Many of these form the first line of defence against asteroids.

Have a pleasant stay

That's it! After these first 3 days of tours, you're free to do whatever you like! The hotel has many activities including "lunar treasure hunt", "Luney Races" with lunar buggies and don't forget to visit Disney's theme park at the Moretus crater. The complete, family holiday destination! 